

The j -formula for the family $\lambda = (2, 2, 1^{n-4})$: the first non-hook structural proof

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Abstract

We extend the May-11 and May-12 structural proofs of the j -formula from hooks $\lambda = (2, 1^{n-2})$ and $\lambda = (3, 1^{n-3})$ to the first non-hook family in the rank-zero regime: $\lambda = (2, 2, 1^{n-4})$ at every $n \geq 6$. The proof tracks the image of the staircase product $R'_{n-1}R'_n$ on V_λ step-by-step in the seminormal basis. After R'_n the image is exactly the $+q$ -eigenspace of T_{n-1} , of dimension $n - 3$, with explicit basis of T_{n-1} 2-block $+q$ -eigenvectors. The first factor (T_1+1) of R'_{n-1} collapses this to a 1-dimensional subspace; subsequent factors shift this subspace through a 4-element “window”, and the final factor $(T_{n-2}+1)$ produces a 1-dimensional image with support on exactly 4 standard tableaux, all in E_1^- . As a corollary, $\Pi^{S_n}|_{V_{(2,2,1^{n-4})}} = 0$ unconditionally for every $n \geq 6$.

This is the first non-hook case where the rank-zero theorem is established structurally. Combined with the hook results of May 11 and May 12, the rank-zero theorem now holds unconditionally for three infinite families in the rank-zero regime.

1 Setup and notation

We work with the Hecke algebra $H_q(S_n)$ over $\mathbb{Q}(q)$, with generators $T_1, \dots, T_{n-1}$ satisfying $T_i^2 = (q-1)T_i + q$ and the standard braid relations. V_λ denotes the seminormal cell module of $H_q(S_n)$ for the partition $\lambda \vdash n$. Its basis $\{v_T\}$ is indexed by standard Young tableaux (SYTs) of shape λ .

The staircase Bott–Samelson element is

$$\Pi^{S_n} = R'_2 R'_3 \cdots R'_n, \quad R'_k = (T_{k-1}+1)(T_{k-2}+1) \cdots (T_1+1).$$

Write $B_j := R'_j R'_{j+1} \cdots R'_n$. $E_i^\pm$ denotes the $\pm$ -eigenspace of T_i (eigenvalues q resp. -1).

1.1 The seminormal basis of $\lambda = (2, 2, 1^{n-4})$

Throughout, fix $\lambda = (2, 2, 1^{n-4})$ for some $n \geq 6$. The shape has length $\ell(\lambda) = n - 2$, total $|\lambda| = n$.

Lemma 1 (SYTs of $\lambda = (2, 2, 1^{n-4})$). *SYTs of shape $\lambda = (2, 2, 1^{n-4})$ are in bijection with pairs (a, c) satisfying $2 \leq a < c \leq n$ and $(a, c) \neq (2, 3)$.*

Proof. For an SYT T of shape λ , write a for the entry at $(1, 2)$ and c for the entry at $(2, 2)$. Necessarily $a > 1$, $c > a$, and the entry b at $(2, 1)$ is the smallest element of $\{2, \dots, n\} \setminus \{a, c\}$. Validity of the SYT requires $b < c$; this fails iff $a = 2$, $c = 3$ (in which case $b = 4 > 3$), and otherwise holds. All remaining entries fill the rest of column 1 in increasing order. $\square$

We write $v_{(a,c)} := v_T$ for T the SYT with second-column entries (a, c) . We refer to “column 1” meaning the leftmost column (positions $(1, 1), (2, 1), \dots, (n-2, 1)$) and “column 2” meaning positions $(1, 2), (2, 2)$.

Lemma 2 (T_i -action on the seminormal basis of λ). For $i \in \{1, \dots, n-1\}$ and SYT $T = T_{(a,c)}$:

- If $i, i+1$ lie in the same row of T : $T_i v_{(a,c)} = q v_{(a,c)}$.
- If $i, i+1$ lie in the same column of T : $T_i v_{(a,c)} = -v_{(a,c)}$.
- Otherwise (different rows and columns), T_i acts as a 2×2 Heefsmits block on $\text{span}(v_T, v_{s_i T})$ where $s_i T$ is the SYT obtained by swapping i and $i+1$. At axial distance $d := c(i+1) - c(i)$ (where $c(r, k) = k - r$), the $+q$ -eigenvector is, up to scalar,

$$u_{T, s_i T}^+ = \alpha_d v_T + \beta_d v_{s_i T}, \quad (T_i + 1)v_T = \kappa_d u_{T, s_i T}^+,$$

with $\alpha_d, \beta_d, \kappa_d \in \mathbb{Q}(q)$ generically nonzero.

Proof. Standard: see e.g. Heefsmits's formula or Murphy's seminormal basis. $\square$

Lemma 3 ($E_1^\pm$ for $\lambda = (2, 2, 1^{n-4})$). T_1 acts diagonally on V_λ (no 2-blocks). Specifically:

$$E_1^+ = \text{span}\{v_{(2,c)} : 4 \leq c \leq n\}, \quad E_1^- = \text{span}\{v_{(a,c)} : 3 \leq a < c \leq n\}.$$

Proof. $T_1 v_{(a,c)}$ depends on the positions of 1 and 2. 1 is always at $(1, 1)$. 2 is at $(1, 2)$ iff $a = 2$ (same row as 1, $T_1 v = qv$), and otherwise at $(2, 1)$ (same column as 1, $T_1 v = -v$). The two cases partition V_λ . $\square$

2 The main theorem

Theorem 4 (j -formula for $\lambda = (2, 2, 1^{n-4})$). For every $n \geq 6$, $\lambda = (2, 2, 1^{n-4}) \vdash n$ has length $\ell = n - 2$ and the j -formula holds with $j = \ell + 1 = n - 1$:

$$B_{n-1}(V_\lambda) := R'_{n-1} R'_n V_\lambda \subseteq E_1^-.$$

Moreover, the image is one-dimensional, with support exactly on the 4 SYTs indexed by

$$\{(n-3, n-2), (n-3, n-1), (n-2, n), (n-1, n)\}.$$

Corollary 5 ($\Pi^{S_n}|_{V_{(2,2,1^{n-4})}} = 0$). $\Pi^{S_n}|_{V_{(2,2,1^{n-4})}} = 0$ unconditionally for every $n \geq 6$.

Proof. By Theorem 4, $B_{n-1} V_\lambda \subseteq E_1^-$. The factorization $\Pi^{S_n} = R'_2 \cdots R'_{n-2} \cdot B_{n-1}$ has R'_{n-2} ending in $(T_1 + 1)$, which annihilates E_1^- . Hence $\Pi^{S_n} V_\lambda = 0$. $\square$

The proof of Theorem 4 occupies Sections 3 (analysis of R'_n) and 4 (analysis of R'_{n-1} on the result of R'_n).

3 Phase A: R'_n surjects onto E_{n-1}^+

For $1 \leq j \leq n-1$, write $W_j^{(n)} := (T_j + 1)(T_{j-1} + 1) \cdots (T_1 + 1) V_\lambda$. Note $W_{n-1}^{(n)} = R'_n V_\lambda$.

Proposition 6 (Phase A endpoint). $W_j^{(n)} = E_j^+$ for every $1 \leq j \leq n-1$. In particular, $\dim W_j^{(n)} = n - 3$.

The proof proceeds via an explicit description of a basis of $W_j^{(n)}$.

Definition 7 (Basis $\mathcal{B}_j$). For $1 \leq j \leq n-1$, define $\mathcal{B}_j$ to be the set of vectors in V_λ given by:

- For each $a \in \{2, \dots, j-1\}$ such that (a, j) is a valid SYT label (i.e., $(a, j) \neq (2, 3)$), the $T_j + q$ -eigenvector $w_j(a)$ in the 2-block $\{v_{(a,j)}, v_{(a,j+1)}\}$. When the 2-block degenerates because $(a, j) = (2, 3)$ is invalid (i.e., $a = 2, j = 3$), we set $w_3(2) := v_{(2,4)}$ (the unique row-2 same-row $+q$ -eigenvector of T_3 in V_λ).
- For each $c \in \{j+2, \dots, n\}$, the $T_j + q$ -eigenvector $w_j(c)$ in the 2-block $\{v_{(j,c)}, v_{(j+1,c)}\}$.

Remark 8 (Block validity). We check that the 2-blocks above genuinely exist. The block $\{v_{(a,j)}, v_{(a,j+1)}\}$ requires T_j to couple them, which happens when j and $j+1$ are in different rows and columns of $T_{(a,j)}$: $j = c$ at $(2, 2)$, while $j+1$ is in column 1 (verifiable when $a < j$ and $(a, j) \neq (2, 3)$). The block $\{v_{(j,c)}, v_{(j+1,c)}\}$ requires $j = a$ at $(1, 2)$ and $j+1$ in column 1, valid when $j+1 < c$. All cases are checked in the inductive proof below.

Lemma 9 (Cardinality of $\mathcal{B}_j$). $|\mathcal{B}_j| = n - 3$ for every $1 \leq j \leq n - 1$.

Proof. Counting: the a -indexed contribute $j - 2$ vectors (one per $a \in \{2, \dots, j-1\}$, with the degenerate case $j = 3, a = 2$ still counted as one vector), and the c -indexed contribute $n - j - 1$ vectors (one per $c \in \{j+2, \dots, n\}$). Total: $(j-2) + (n-j-1) = n-3$. For $j = 1, 2$, the a -indexed range is empty, and the c -indexed range $\{j+2, \dots, n\}$ has $n-j-1$ elements intersected with the constraint (j, c) or $(j+1, c)$ are valid SYT labels — we verify case-by-case below.

For $j = 1$: $\mathcal{B}_1 = \{w_1(c) : c \in \{3, \dots, n\}\}$ where $w_1(c)$ is supposed to be the $+q$ -eigenvector in the block $\{v_{(1,c)}, v_{(2,c)}\}$. Since $(1, c)$ is not a valid SYT label, there is no 2-block; instead T_1 acts diagonally and $w_1(c) := v_{(2,c)}$ for $c \geq 4$ (with $c = 3$ excluded as $(2, 3)$ is not a valid SYT label). This gives $|\mathcal{B}_1| = n - 3$.

For $j = 2$: $\mathcal{B}_2 = \{w_2(c) : c \in \{4, \dots, n\}\}$. $|\mathcal{B}_2| = n - 3$. □

Lemma 10 (Inductive transition $\mathcal{B}_{j-1} \rightarrow \mathcal{B}_j$). For $2 \leq j \leq n - 1$, the map $(T_j + 1)$ takes $\mathcal{B}_{j-1}$ bijectively (up to nonzero scalars) to $\mathcal{B}_j$, via the transformation:

- $w_{j-1}(a) \mapsto (\text{nonzero scalar}) \cdot w_j(a)$ for $a \in \{2, \dots, j-2\}$ (those a -indexed vectors of $\mathcal{B}_{j-1}$ for which $a \leq j-2$).
- $w_{j-1}(c = j+1) \mapsto (\text{nonzero scalar}) \cdot w_j(a = j-1)$ (this maps the c -indexed vector with $c = j$ in $\mathcal{B}_{j-1}$, i.e., $c = (j-1) + 2$, to the new a -indexed vector at $a = j-1$ in $\mathcal{B}_j$).
- $w_{j-1}(c) \mapsto (\text{nonzero scalar}) \cdot w_j(c)$ for $c \in \{j+2, \dots, n\}$.

In particular, $\mathcal{B}_j$ is a basis of $W_j^{(n)}$, so $\dim W_j^{(n)} = n - 3 = \dim E_j^+$, hence $W_j^{(n)} = E_j^+$.

Proof. We compute $(T_j + 1)$ applied to each basis vector of $\mathcal{B}_{j-1}$ in the seminormal basis.

(I) $w_{j-1}(a)$ for $a \in \{2, \dots, j-2\}$. By Definition 7 (and assuming $a < j-1$, $(a, j-1) \neq (2, 3)$), $w_{j-1}(a) = \alpha v_{(a,j-1)} + \beta v_{(a,j)}$.

Compute $(T_j + 1)v_{(a,j-1)}$. In $T_{(a,j-1)}$ with $a \leq j-2$: a at $(1, 2)$, $c = j-1$ at $(2, 2)$. j and $j+1$ are both in column 1 (since neither equals a nor c). Counting: column 1 entries are $\{1\} \cup \{2, \dots, n\} \setminus \{a, j-1\}$; elements $\leq j-1$ in this set total $j-2$, so position of j is row $j-1$ if j is the $(j-1)$ th smallest. Direct verification: column 1 entries less than j are $\{1, 2, \dots, j-1\} \setminus \{a, j-1\}$, totaling $j-3$ elements, plus $\{1\}$ gives $j-2$. Therefore j is at row $j-1$, position $(j-1, 1)$. Likewise $j+1$ at $(j, 1)$. Same column adjacent: $T_j v = -v$, so $(T_j + 1)v_{(a,j-1)} = 0$.

Compute $(T_j + 1)v_{(a,j)}$. In $T_{(a,j)}$: a at $(1, 2)$, $c = j$ at $(2, 2)$. $j+1$ is in column 1 (not equal to a or c). Counting: column 1 entries less than $j+1$ are $\{1, \dots, j\} \setminus \{a, j\}$, totaling $j-2$; so $j+1$ is at row $j-1$, position $(j-1, 1)$, content $-(j-2)$. The cell of j is $(2, 2)$, content 0. Different positions.

T_j -block partner: swap $j, j+1$ moves $j+1$ to $(2, 2)$, j to $(j-1, 1)$; new SYT label $(a, j+1)$. Axial distance $d = -(j-2) - 0 = -(j-2)$, so $(T_j+1)v_{(a,j)} = \kappa u_{T_{(a,j)}, T_{(a,j+1)}}^+ = \kappa(\alpha'v_{(a,j)} + \beta'v_{(a,j+1)})$ for nonzero α', β', κ depending on j, n, q .

Combining: $(T_j+1)w_{j-1}(a) = \beta\kappa(\alpha'v_{(a,j)} + \beta'v_{(a,j+1)})$, which is a nonzero scalar multiple of $w_j(a)$. $\checkmark$

(II) $w_{j-1}(c = j+1)$. By Definition 7, $w_{j-1}(c = j+1) = \alpha v_{(j-1,j+1)} + \beta v_{(j,j+1)}$ (T_{j-1} -block).

Compute $(T_j+1)v_{(j-1,j+1)}$. In $T_{(j-1,j+1)}$: $a = j-1$ at $(1, 2)$, $c = j+1$ at $(2, 2)$. j in column 1 (not equal to a or c). Position: j has $j-2$ smaller elements in column 1 ($\{1, 2, \dots, j-1\} \setminus \{j-1\} = \{1, \dots, j-2\}$), so j at $(j-1, 1)$, content $-(j-2)$. Different from $(2, 2)$ where $j+1$ sits. T_j -block partner: swap $j, j+1$ moves $j+1$ to $(j-1, 1)$, j to $(2, 2)$; new SYT $T_{(j-1,j)}$. Axial distance $0 - (-(j-2)) = j-2$, so $(T_j+1)v_{(j-1,j+1)} \propto u^+ = \alpha'v_{(j-1,j)} + \beta'v_{(j-1,j+1)}$ (a T_j 2-block $+q$ -eigenvector with support on $\{(j-1, j), (j-1, j+1)\}$).

Compute $(T_j+1)v_{(j,j+1)}$. $a = j, c = j+1$: same column 2. $T_j v = -v$, so $(T_j+1)v = 0$.

Combining: $(T_j+1)w_{j-1}(c = j+1) = \alpha\kappa(\alpha'v_{(j-1,j)} + \beta'v_{(j-1,j+1)})$, a nonzero scalar multiple of $w_j(a = j-1)$. $\checkmark$

(III) $w_{j-1}(c)$ for $c \in \{j+2, \dots, n\}$. $w_{j-1}(c) = \alpha v_{(j-1,c)} + \beta v_{(j,c)}$.

Compute $(T_j+1)v_{(j-1,c)}$. $a = j-1, c \geq j+2$. j and $j+1$ both in column 1 (since neither equals a nor c). Adjacent rows. $T_j v = -v$, so $(T_j+1)v = 0$.

Compute $(T_j+1)v_{(j,c)}$. $a = j$ at $(1, 2)$, content 1. $j+1$ in column 1 (not equal to a , and $c \geq j+2 \neq j+1$). Counting: column 1 entries less than $j+1$ are $\{1, 2, \dots, j\} \setminus \{j\} = \{1, \dots, j-1\}$, totaling $j-1$. So $j+1$ is at $(j, 1)$, content $1-j$. Different from $(1, 2)$. T_j -block partner: swap moves $j+1$ to $(1, 2)$, j to $(j, 1)$; new SYT $T_{(j+1,c)}$. Axial distance $1-j-1 = -j$. $(T_j+1)v_{(j,c)} \propto u^+ = \alpha'v_{(j,c)} + \beta'v_{(j+1,c)}$, a T_j 2-block $+q$ -eigenvector with support $\{(j, c), (j+1, c)\}$.

Combining: $(T_j+1)w_{j-1}(c) = \beta\kappa(\alpha'v_{(j,c)} + \beta'v_{(j+1,c)})$, a nonzero scalar multiple of $w_j(c)$. $\checkmark$

Conclusion. (T_j+1) takes $\mathcal{B}_{j-1}$ to a set of $n-3$ vectors, each a nonzero scalar multiple of a distinct element of $\mathcal{B}_j$. The $n-3$ vectors of $\mathcal{B}_j$ have pairwise distinct supports (by inspection: the a -indexed have support $\{(a, j), (a, j+1)\}$ with $a \leq j-1$, the c -indexed have support $\{(j, c), (j+1, c)\}$ with $c \geq j+2$, and the supports are pairwise disjoint). Hence $\mathcal{B}_j$ is linearly independent, and spans $W_j^{(n)}$.

The image $W_j^{(n)} \subseteq E_j^+$ since $(T_j+1) = (1+q)\pi_j^+$ where π_j^+ is the projection onto E_j^+ along E_j^- . Comparing dimensions: $|\mathcal{B}_j| = n-3$ and we'll show $\dim E_j^+ \leq n-3$ by counting (Lemma 11 below), giving equality. $\square$

Lemma 11 (Dimension of E_j^+). *For each $1 \leq j \leq n-1$, $\dim E_j^+ = n-3$.*

Proof. Since T_j and $T_{j'}$ are conjugate in $H_q(S_n)$ (any two simple reflections are conjugate via a sequence of braid moves), their traces on the irreducible V_λ are equal. Hence $\text{tr}(T_j|_{V_\lambda})$ is a constant in j . As T_j is diagonalizable on V_λ with eigenvalues $\{q, -1\}$,

$$\text{tr}(T_j|_{V_\lambda}) = q \cdot \dim E_j^+ - \dim E_j^-, \quad \dim E_j^+ + \dim E_j^- = \dim V_\lambda = \frac{n(n-3)}{2}.$$

Solving, $\dim E_j^+ = (\text{tr}(T_j) + \dim V_\lambda)/(q+1)$, which is independent of j .

Compute at $j = 1$ using Lemma 3: $\dim E_1^+ = n-3$ (the SYTs $v_{(2,c)}$ for $4 \leq c \leq n$). Hence $\dim E_j^+ = n-3$ for all j . $\square$

Proof of Proposition 6. Combining Lemmas 9, 10, 11: $W_j^{(n)} \subseteq E_j^+$, and both have dimension $n-3$, so $W_j^{(n)} = E_j^+$. $\square$

Corollary 12 (W_n described). $W_n := R'_n V_\lambda = E_{n-1}^+$ has basis $\mathcal{B}_{n-1} = \{w_{n-1}(a) : a \in \{2, \dots, n-2\}\}$ where $w_{n-1}(a) = \alpha v_{(a,n-1)} + \beta v_{(a,n)}$ is the $T_{n-1} + q$ -eigenvector in the 2-block $\{v_{(a,n-1)}, v_{(a,n)}\}$.

Proof. At $j = n-1$: a -indexed range is $\{2, \dots, n-2\}$ ($n-3$ elements), c -indexed range $\{n+1, \dots, n\}$ is empty. Each $w_{n-1}(a)$ is a T_{n-1} 2-block $+q$ -eigenvector by construction. $\square$

4 Phase B: collapse to E_1^- via R'_{n-1}

We now apply $R'_{n-1} = (T_{n-2}+1) \cdots (T_1+1)$ to W_n . Define $\widetilde{W}_j := (T_j+1) \cdots (T_1+1) W_n$ for $1 \leq j \leq n-2$.

Proposition 13 (Step 1: collapse to dim 1). $\widetilde{W}_1 = \text{span}(w_{n-1}(2))$, 1-dimensional.

Proof. By Corollary 13, W_n has basis $\{w_{n-1}(a) : 2 \leq a \leq n-2\}$ with $w_{n-1}(a) = \alpha v_{(a,n-1)} + \beta v_{(a,n)}$.

Case $a = 2$. Both $v_{(2,n-1)}$ and $v_{(2,n)}$ have $2 \in \{a\}$ in row 1, so by Lemma 3 they lie in E_1^+ . Hence $T_1 w_{n-1}(2) = q w_{n-1}(2)$, so $(T_1+1)w_{n-1}(2) = (1+q)w_{n-1}(2)$. Survives.

Case $a \geq 3$. Both $v_{(a,n-1)}$ and $v_{(a,n)}$ have $a \geq 3$, so 2 is in column 1, hence in E_1^- (Lemma 3). $(T_1+1)w_{n-1}(a) = 0$. Killed.

Therefore $\widetilde{W}_1 = \text{span}(w_{n-1}(2))$. $\square$

Proposition 14 (Steps 2 through $n-3$: shifting window). For $2 \leq j \leq n-3$,

$$\widetilde{W}_j = \text{span}(\widetilde{u}_j), \quad \widetilde{u}_j = A_j v_{(j,n-1)} + B_j v_{(j+1,n-1)} + C_j v_{(j,n)} + D_j v_{(j+1,n)},$$

with $A_j, B_j, C_j, D_j \in \mathbb{Q}(q)$ all nonzero.

Proof. By induction on j .

Base case $j = 2$. Apply (T_2+1) to $\widetilde{W}_1 = \text{span}(w_{n-1}(2))$ where $w_{n-1}(2) = \alpha v_{(2,n-1)} + \beta v_{(2,n)}$.

For $c \in \{n-1, n\}$ ($c \geq 4$): in $T_{(2,c)}$, $a = 2$ at $(1, 2)$, $b = 3$ at $(2, 1)$ (smallest of $\{2, \dots, n\} \setminus \{2, c\}$ is 3). T_2 couples $v_{(2,c)}$ with $v_{(3,c)}$ (axial distance -2), so $(T_2+1)v_{(2,c)} = \kappa(\alpha_2 v_{(2,c)} + \beta_2 v_{(3,c)})$ with $\alpha_2, \beta_2, \kappa$ nonzero.

Hence $(T_2+1)w_{n-1}(2) = \kappa(\alpha\alpha_2 v_{(2,n-1)} + \alpha\beta_2 v_{(3,n-1)} + \beta\alpha_2 v_{(2,n)} + \beta\beta_2 v_{(3,n)})$, all four coefficients nonzero. This matches the form $\widetilde{u}_2$ with $j = 2$, supported on $\{(2, n-1), (3, n-1), (2, n), (3, n)\}$.

Inductive step $j-1 \rightarrow j$. Assume $\widetilde{W}_{j-1}$ is the span of $\widetilde{u}_{j-1}$ supported on $\{(j-1, n-1), (j, n-1), (j-1, n), (j, n)\}$ with all four coefficients nonzero. Apply (T_j+1) for $2 \leq j \leq n-3$.

(i) $v_{(j-1,c)}$ for $c \in \{n-1, n\}$. $a = j-1, c \in \{n-1, n\}$. Since $j \leq n-3 < n-1 \leq c$, j and $j+1$ are both $\leq n-2 < n-1$, so neither equals $a = j-1$ nor c . Hence $j, j+1$ are in column 1.

Position counting: column 1 entries are $\{1\} \cup \{2, \dots, n\} \setminus \{j-1, c\}$, sorted. Elements less than j in column 1 are $\{1, \dots, j-1\} \setminus \{j-1\} = \{1, \dots, j-2\}$, totaling $j-1$ elements. So j is at row j , position $(j, 1)$, content $1-j$. Elements less than $j+1$ in column 1 are $\{1, \dots, j\} \setminus \{j-1\} = \{1, \dots, j-2, j\}$, totaling $j-1$. Wait: $\{1, \dots, j\}$ has j elements, removing one gives $j-1$ if that one is in the set (and $j-1 \in \{1, \dots, j\}$ iff $j \geq 2$). So $j-1$ elements. Hmm, this would put $j+1$ at row j , but j is already there. Let me recount.

Number of column-1 elements strictly less than j : we need elements of $\{1\} \cup \{2, \dots, n\} \setminus \{j-1, c\}$ that are strictly less than j . Elements: $\{1, 2, \dots, j-1\} \setminus \{j-1\} = \{1, \dots, j-2\}$ (since $c > j$), total $j-2$ elements (for $j \geq 2$). So j is at row $j-1$, position $(j-1, 1)$, content $-(j-2) = 2-j$.

Elements strictly less than $j+1$ in column 1: $\{1, \dots, j\} \setminus \{j-1\} = \{1, \dots, j-2, j\}$, total $j-1$ elements. So $j+1$ is at row j , position $(j, 1)$, content $1-j$.

So j at $(j-1, 1)$, $j+1$ at $(j, 1)$: same column adjacent. $T_j v_{(j-1, c)} = -v_{(j-1, c)}$ for $c \in \{n-1, n\}$. Hence $(T_j+1)v_{(j-1, c)} = 0$.

(ii) $v_{(j, c)}$ for $c \in \{n-1, n\}$. $a = j$ at $(1, 2)$, content 1. $j+1$ in column 1 (not equal to $a = j$ or c , since $j+1 \leq n-2 < n-1 \leq c$).

Position of $j+1$: number of column-1 entries less than $j+1$ is $|\{1\} \cup \{2, \dots, j\} \setminus \{j, c\}| = |\{1\} \cup \{2, \dots, j-1\}| = j-1$. So $j+1$ at row j , position $(j, 1)$, content $1-j$.

Different from $(1, 2)$. T_j -block partner: swap $j, j+1$ moves $j+1$ to $(1, 2)$, j to $(j, 1)$; new SYT $T_{(j+1, c)}$. Axial distance $1-j-1 = -j$. $(T_j+1)v_{(j, c)} = \kappa_j(\alpha' v_{(j, c)} + \beta' v_{(j+1, c)})$ with $\kappa_j, \alpha', \beta'$ nonzero.

Combining.

$$\begin{aligned} (T_j+1)\tilde{u}_{j-1} &= (T_j+1)(A_{j-1}v_{(j-1, n-1)} + B_{j-1}v_{(j, n-1)} + C_{j-1}v_{(j-1, n)} + D_{j-1}v_{(j, n)}) \\ &= 0 + B_{j-1}\kappa_j(\alpha' v_{(j, n-1)} + \beta' v_{(j+1, n-1)}) + 0 + D_{j-1}\kappa_j(\alpha' v_{(j, n)} + \beta' v_{(j+1, n)}) \\ &= \kappa_j(B_{j-1}\alpha' v_{(j, n-1)} + B_{j-1}\beta' v_{(j+1, n-1)} + D_{j-1}\alpha' v_{(j, n)} + D_{j-1}\beta' v_{(j+1, n)}), \end{aligned}$$

which has all four coefficients nonzero ($B_{j-1}, D_{j-1}, \alpha', \beta', \kappa_j$ all nonzero). This is exactly $\tilde{u}_j$ supported on $\{(j, n-1), (j+1, n-1), (j, n), (j+1, n)\}$. $\square$

Proposition 15 (Step $n-2$: terminal shape). $\widetilde{W}_{n-2}$ is 1-dimensional, supported on the 4 SYTs

$$\{(n-3, n-2), (n-3, n-1), (n-2, n), (n-1, n)\},$$

and contained in E_1^- .

Proof. Apply $(T_{n-2}+1)$ to $\tilde{u}_{n-3}$ supported on $\{(n-3, n-1), (n-2, n-1), (n-3, n), (n-2, n)\}$ (by Prop. 15 with $j = n-3$, swapping the roles): write $\tilde{u}_{n-3} = Av_{(n-3, n-1)} + Bv_{(n-2, n-1)} + Cv_{(n-3, n)} + Dv_{(n-2, n)}$ with $A, B, C, D \neq 0$.

(i) $v_{(n-3, n-1)}$. $a = n-3$ at $(1, 2)$, $c = n-1$ at $(2, 2)$. $n-2$: not in $\{a, c\}$. Position: column 1 entries are $\{1\} \cup \{2, \dots, n\} \setminus \{n-3, n-1\} = \{1, 2, \dots, n-4, n-2, n\}$. Elements less than $n-2$ in this list: $\{1, 2, \dots, n-4\}$, totaling $n-4$. So $n-2$ at row $n-3$, position $(n-3, 1)$, content $-(n-4)$. $n-1$ at $(2, 2)$, content 0. Different. T_{n-2} -block partner: swap $n-2, n-1$ moves $n-1$ to $(n-3, 1)$, $n-2$ to $(2, 2)$; new SYT $T_{(n-3, n-2)}$. Axial distance $0 - (-(n-4)) = n-4$. $(T_{n-2}+1)v_{(n-3, n-1)} = \kappa(\alpha' v_{(n-3, n-1)} + \beta' v_{(n-3, n-2)})$ with $\kappa, \alpha', \beta' \neq 0$.

(ii) $v_{(n-2, n-1)}$. $a = n-2, c = n-1$: same column 2 ($a+1=c$). $T_{n-2}v = -v$. $(T_{n-2}+1)v = 0$.

(iii) $v_{(n-3, n)}$. $a = n-3, c = n$. $n-2, n-1$: both not in $\{a, c\}$, so both in column 1. Column 1 entries: $\{1\} \cup \{2, \dots, n\} \setminus \{n-3, n\} = \{1, 2, \dots, n-4, n-2, n-1\}$. $n-2$ at row $n-3$, content $-(n-4)$; $n-1$ at row $n-2$, content $-(n-3)$. Same column adjacent. $T_{n-2}v = -v$. $(T_{n-2}+1)v = 0$.

(iv) $v_{(n-2, n)}$. $a = n-2$ at $(1, 2)$, content 1. $c = n$ at $(2, 2)$. $n-1$: column 1 (not equal to a or c). Column 1: $\{1\} \cup \{2, \dots, n\} \setminus \{n-2, n\} = \{1, 2, \dots, n-3, n-1\}$. $n-1$ at row $n-2$, position $(n-2, 1)$, content $-(n-3)$. Different from $(1, 2)$. T_{n-2} -block partner: swap $n-2, n-1$ moves $n-1$ to $(1, 2)$, $n-2$ to $(n-2, 1)$; new SYT $T_{(n-1, n)}$. Axial distance $-(n-3) - 1 = -(n-2)$. $(T_{n-2}+1)v_{(n-2, n)} = \kappa'(\alpha'' v_{(n-2, n)} + \beta'' v_{(n-1, n)})$ with $\kappa', \alpha'', \beta'' \neq 0$.

Combining.

$$\begin{aligned} (T_{n-2}+1)\tilde{u}_{n-3} &= A\kappa(\alpha' v_{(n-3, n-1)} + \beta' v_{(n-3, n-2)}) + 0 + 0 \\ &\quad + D\kappa'(\alpha'' v_{(n-2, n)} + \beta'' v_{(n-1, n)}). \end{aligned}$$

Four nonzero coefficients (since $A, D, \kappa, \kappa', \alpha', \alpha'', \beta', \beta'' \neq 0$); supports as claimed.

E_1^- containment. Every support has minimum element $\geq n-3 \geq 3$ (for $n \geq 6$). By Lemma 3, $2 \notin S$ implies $v_S \in E_1^-$. Hence $\tilde{u}_{n-2} \in E_1^-$. $\square$

Proof of Theorem 4. Combine Proposition 6 (Phase A endpoint $W_n = E_{n-1}^+$, $\dim n-3$), Proposition 14 (rank collapses to 1), Proposition 15 (shifting window), Proposition 16 (terminal step). $B_{n-1}V_\lambda = R'_{n-1}R'_nV_\lambda = \tilde{W}_{n-2}$ is one-dimensional, supported on $\{(n-3, n-2), (n-3, n-1), (n-2, n), (n-1, n)\}$, and contained in E_1^- . $\square$

5 Computational verification

Theorem 16 (Verification at $n = 6, \dots, 9$). *Theorem 4 and the intermediate steps (Propositions 6, 14, 15, 16) hold computationally at $n = 6, 7, 8, 9$.*

Verification. SymPy at $q = 7 \in \mathbb{Q}$ (generic for the seminormal basis). At each step, the rank, support pattern, and E_1^- containment match the predicted values. At $n = 9$, $\tilde{W}_{n-2}$ is supported on $\{(6, 7), (6, 8), (7, 9), (8, 9)\}$, all with min element $6 \geq 3$, hence in E_1^- . Script: `~/projects/scratch/2026-05-13-no`. The script verifies:

- Phase A: at every $j \in \{1, \dots, n-1\}$, rank of $W_j^{(n)}$ is exactly $n-3$, and the support matches the explicit prediction from $\mathcal{B}_j$.
- Phase B: at every $j \in \{1, \dots, n-2\}$, rank of $\tilde{W}_j$ is exactly 1, and the support matches the prediction. E_1^- containment confirmed for $j \geq 3$ (and specifically at $j = n-2$).

$\square$

6 Discussion

6.1 Comparison with the hook results

The structural pattern — Phase A landing in E_{n-1}^+ , Phase B collapsing to 1-dimensional via (T_1+1) , then shifting through a window, then a terminal step — closely parallels the May-12 proof for the hook $\lambda = (3, 1^{n-3})$ (writeup `2026-05-12-hook-j-formula.tex`).

The key difference is the terminal support pattern:

- Hook $(3, 1^{n-3})$: 5 SYTs in support, namely $\{(n-3, n-2), (n-3, n-1), (n-2, n-1), (n-2, n), (n-1, n)\}$.
- Non-hook $(2, 2, 1^{n-4})$: 4 SYTs in support, namely $\{(n-3, n-2), (n-3, n-1), (n-2, n), (n-1, n)\}$.

The non-hook case has *one fewer* SYT in the support: namely $(n-2, n-1)$ is excluded. This is because in $\lambda = (2, 2, 1^{n-4})$ the SYT $T_{(n-2, n-1)}$ has $(a, c) = (n-2, n-1)$ in column 2 together — a same-column-2 configuration for T_{n-2} , which $(T_{n-2}+1)$ kills. In the hook case, the analogous SYT has $\{n-2, n-1\}$ both in row 1, which $(T_{n-2}+1)$ acts on as $(1+q)$ (same-row), hence the SYT $(n-2, n-1)$ *survives* and contributes to the support.

6.2 Cardinality of supports

We have now seen the following support sizes for $B_{\ell+1}V_\lambda$ in the rank-zero regime:

- $\lambda = (2, 1^{n-2})$ ($k = 2$ hook): support size 2.

- $\lambda = (3, 1^{n-3})$ ($k = 3$ hook): support size $5 = C_3$.
- $\lambda = (2, 2, 1^{n-4})$ (current paper): support size 4.

The hook support sizes match Catalan numbers C_k (2026-05-12-hook-j-formula.tex Conj. 3.1). The non-hook support size 4 does not directly match any obvious Catalan-type count. This raises the question:

Conjecture 17 (Support size for general non-hooks). *For $\lambda \vdash n$ in the rank-zero regime, the support size of the unique (up to scalar) spanning vector of $B_{\ell+1}V_\lambda$ is governed by a $\kappa(\lambda)$ -tuple of Catalan-like numbers, one per length-preserving corner of λ . For $\lambda = (2, 2, 1^{n-4})$ (one corner at row 2): the relevant “Catalan-like” number is $\binom{2}{1} \cdot C_2 = 2 \cdot 2 = 4$, matching our count.*

The factor $\binom{2}{1}$ would account for the choice of column 1 vs. column 2 at the corner: a non-hook corner allows “two ways out” compared to a hook. This is purely speculative; verifying the formula on more non-hook cases is the natural next step.

6.3 Coverage of the rank-zero theorem

Together with the May-11 and May-12 proofs, the rank-zero theorem $\Pi^{S_n}|_{V_\lambda} = 0$ is now *structurally proven* for the following infinite families:

- $\lambda = (1^n)$ (sign rep): trivial, since Π^{S_n} contains (T_i+1) for every i , and (T_i+1) acts as 0 on the sign rep.
- $\lambda = (2, 1^{n-2})$ (May 11): hook $k = 2$ family.
- $\lambda = (3, 1^{n-3})$ for $n \geq 6$ (May 12): hook $k = 3$ family.
- $\lambda = (2, 2, 1^{n-4})$ for $n \geq 6$ (current paper): first non-hook family.

The remaining cases in the rank-zero regime (e.g., $(4, 1^{n-4})$ hook, $(3, 2, 1^{n-5})$, $(2, 2, 2, 1^{n-6})$, etc.) remain open structurally.

7 Honest assessment

What this paper achieves.

1. Theorem 4 is proven structurally for every $n \geq 6$. The proof is a step-by-step computation in the seminormal basis, with κ and similar nonzero scalars tracked but not computed explicitly.
2. Corollary 5: $\Pi^{S_n}|_{V_{(2,2,1^{n-4})}} = 0$ for all $n \geq 6$, the first non-hook structural rank-zero result.
3. The terminal support has size 4, not $C_3 = 5$ as in the hook $(3, 1^{n-3})$ case. The structural reason — one SYT $(n-2, n-1)$ is excluded due to the same-column-2 killing — is identified explicitly.
4. Conjecture 18: a tentative Catalan-product formula for support sizes of non-hooks, with $\binom{k}{j} \cdot C_j$ as the relevant counting unit per length-preserving corner.

What this paper does *not* achieve.

1. Conjecture 18 is verified at $n \leq 9$ for $\lambda = (2, 2, 1^{n-4})$ but unverified for other non-hooks.
2. The relationship between the support of $B_{\ell+1}V_\lambda$ and the SYTs of $\lambda^{>1} = (1, 1)$ (the partition obtained by removing column 1) is suggestive but not made precise.

3. Phase A of the proof requires a delicate inductive argument (Lemma 10) that handles three sub-cases. A more “categorical” or “functorial” version, possibly via branching $V_\lambda \downarrow_{S_{n-1}}$ or via an interpolation/limit argument, would be aesthetically more satisfying.

Status. The structural rank-zero theorem now covers two hook families and one non-hook family. The path forward is either (a) extending to more non-hook families ($\lambda = (2, 2, 2, 1^{n-6})$ being the most natural test case), or (b) finding a uniform proof covering all λ in the rank-zero regime. The latter likely requires a deeper invariant (representation-theoretic or geometric) that we have not yet identified.